Breast Ultrasound Image Classification using VGG16

1. Introduction

In this project, a deep learning model is used to classify breast ultrasound images into three categories:

- Benign
- Malignant
- Normal

The dataset used is the **Breast Ultrasound Images Dataset (BUSI)**. A pretrained deep learning model called **VGG16** is used to perform the classification.

2. Dataset Description

The BUSI dataset contains ultrasound images of breast tissue. The images belong to three classes:

Class	**Meaning**
Benign	Non-cancer tumor
Malignant	Cancer tumor
Normal	No tumor

The dataset also contains mask images that highlight tumor areas. Since this project focuses on classification, the mask images were removed during preprocessing.

The dataset also has **class imbalance**, meaning the number of images in each class is different.

Class	**Approximate Images**
Benign	437
Malignant	210
Normal	133

Because benign images are more in number, the model may sometimes predict benign more often.

3. Methodology

Data Preprocessing

Before training the model, the images were prepared using the following steps:

- Mask images containing **"_mask"** were removed.
- Images were resized to **224 × 224 pixels by using** OpenCV library.
- Pixel values were normalized by dividing by **255**, as it is easy for deep learning model (VGG16) to train if all the values are between 0 to 1.

Dataset Splitting

The dataset was divided into three parts:

Dataset	**Percentage**
Training	70%
Validation	15%
Testing	15%

The **training set** is used to train the model.
The **validation set** is used to check the model during training.
The **test set** is used for final evaluation.

Keeping these datasets separate prevents **data leakage**, meaning the model does not see the test data during training.

Model Used

In this project, the **VGG16 pretrained model** is used. The convolution layers of VGG16 are used to extract important features from the ultrasound images, such as edges and patterns.

I have **frozen** these convolutional layers, meaning their weights were not changed during training. And I have removed the classification layers of VGG16 as they are to classify ImageNet dataset.

New layers were added at the end for classification:

- Flatten layer
- Dense layer
- Softmax output layer

The final output layer(Dense layer) has **3 neurons**, representing:

- Benign
- Malignant
- Normal

4. Training Process

The model was trained using the **Adam optimizer** and **categorical cross-entropy loss function.**

Training was done for different numbers of epochs seperately :

- 10 epochs
- 20 epochs
- 30 epochs

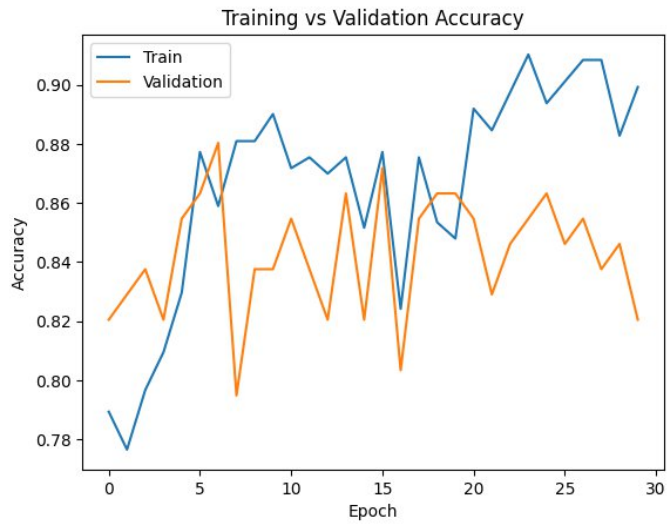
This was done to observe how the model performance changes with more training.

5. Training Graphs

Training vs Validation Accuracy

The accuracy graph shows how the model improves during training.

The training accuracy increases from around **78% to about 90%** as the number of epochs increases. This means the model is learning useful features from the images.The validation accuracy stays between **82% and 86%**, which shows that the model performs well on unseen validation data.

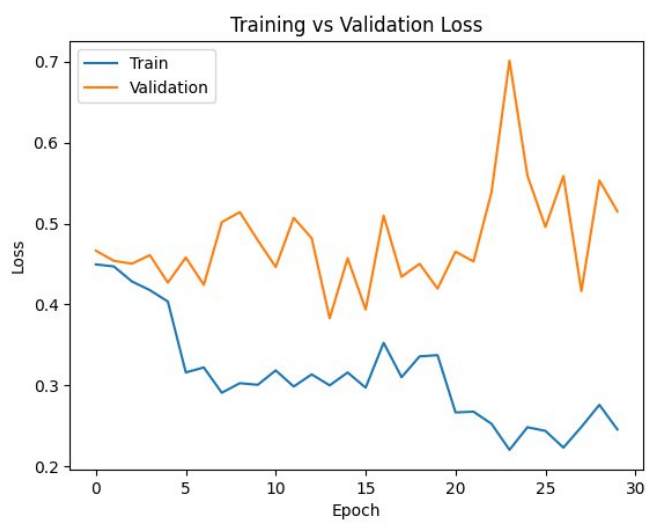


Training vs Validation Loss

The loss graph shows how the model error changes during training.

The training loss decreases steadily, which means the model is improving its predictions.

The validation loss fluctuates slightly. This may happen because the dataset is small and has class imbalance.



6. Results

Results for 10 Epochs

Test Accuracy: **83.05%**

```
test_loss, test_acc = model.evaluate(X_test,y_test)

print("Test Accuracy:",test_acc)

4/4 ━━━━━━━━━━━━━━━━━━━━ 83s 18s/step - accuracy: 0.8305 - loss: 0.5251
Test Accuracy: 0.8305084705352783
```

	precision	recall	f1-score	support
Benign	0.84	0.92	0.88	71
Malignant	0.80	0.80	0.80	30
Normal	0.82	0.53	0.64	17
accuracy			0.83	118
macro avg	0.82	0.75	0.77	118
weighted avg	0.83	0.83	0.82	118

Results for 20 Epochs

Test Accuracy: **83.89%**

```
test_loss, test_acc = model.evaluate(X_test,y_test)

print("Test Accuracy:",test_acc)

4/4 ━━━━━━━━━━━━━━━━━━━━ 63s 15s/step - accuracy: 0.8390 - loss: 0.4608
Test Accuracy: 0.8389830589294434
```

	precision	recall	f1-score	support
Benign	0.84	0.92	0.88	71
Malignant	0.80	0.80	0.80	30
Normal	0.91	0.59	0.71	17
accuracy			0.84	118
macro avg	0.85	0.77	0.80	118
weighted avg	0.84	0.84	0.83	118

Results for 30 Epochs

Test Accuracy: **84.7%**

```
test_loss, test_acc = model.evaluate(X_test,y_test)

print("Test Accuracy:",test_acc)

4/4 ━━━━━━━━━━━━━━━━━━━━ 64s 15s/step - accuracy: 0.8475 - loss: 0.5981
Test Accuracy: 0.8474576473236084
```

	precision	recall	f1-score	support
Benign	0.84	0.92	0.88	71
Malignant	0.80	0.80	0.80	30
Normal	0.91	0.59	0.71	17
accuracy			0.84	118
macro avg	0.85	0.77	0.80	118
weighted avg	0.84	0.84	0.83	118

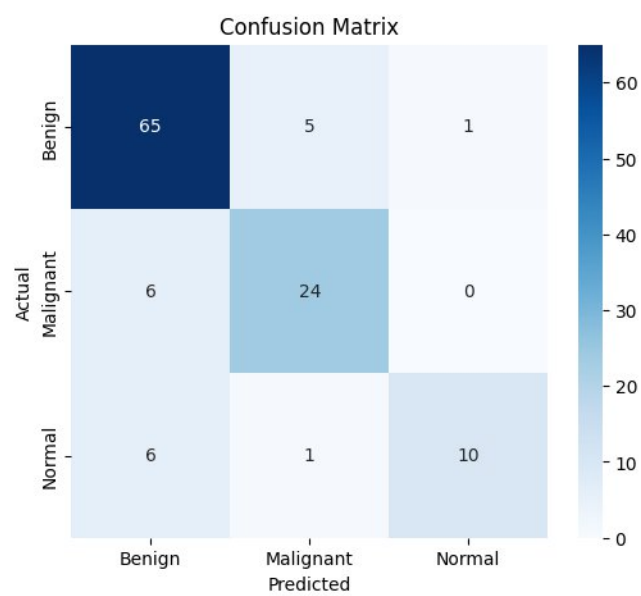
The best performance was obtained with **30 epochs**, where the model achieved around **85% accuracy**.

7. Confusion Matrix

Confusion Matrix Analysis

The confusion matrix shows how many images were correctly and incorrectly classified by the model.

Out of **71 benign images**, **65 were correctly classified**, while 5 were predicted as malignant and 1 as normal.
Out of **30 malignant images**, **24 were correctly classified**, while 6 were predicted as benign.
Out of **17 normal images**, **10 were correctly classified**, while 6 were predicted as benign and 1 as malignant.



Overall, the model performs well for **benign and malignant classes**, while the **normal class has slightly lower accuracy**, possibly due to the smaller number of normal samples in the dataset.

8. Evaluation Metrics

The model performance was measured using the following metrics:

- **Accuracy :**Shows the overall percentage of correct predictions. Final accuracy - **84.7%**
- **Precision** : Measures how many predicted cases were actually correct.
- **Recall** : Measures how many actual cases were correctly detected.
- **F1 Score** : A balance between precision and recall.

9. Conclusion

In this project, a deep learning model using VGG16 transfer learning was used to classify breast ultrasound images into three classes: benign, malignant, and normal.

The dataset was preprocessed by removing the mask images, resizing the images to the required size, and normalizing the pixel values. The dataset was then divided into training, validation, and testing sets to make sure the model does not see the test data during training.

The model performed best when trained for 30 epochs, achieving an accuracy of about 85%. These results show that deep learning models can help in automatically classifying breast ultrasound images.